

Hannah Binney

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EDUCATION

University of Washington

Ph.D., Physics

Dec. 2022

Seattle, WA

- Thesis titled “Precession frequency analysis for Run-2 and Run-3 of the Muon g-2 experiment.”

University of Washington

M.S., Physics

Oct. 2017

Seattle, WA

University of Chicago

B.A., Physics with Honors; Minor in English & Creative Writing

Jun. 2016

Chicago, IL

- Thesis titled “13 TeV studies for vector-boson-fusion production of Higgs decaying to bb at ATLAS.”

SKILLS

- | | | |
|---|--|------------------------------|
| ■ Python (matplotlib, numpy, astropy, astroquery) | ■ C++ (ROOT and <i>art</i> frameworks) | ■ High performance computing |
| ■ Matlab | ■ Mathematica | ■ Particle physics detectors |
| ■ LaTeX | ■ Unix | |

EXPERIENCE

Massachusetts Institute of Technology

Postdoctoral Research Associate, Advisor: Joe Formaggio

July 2024–Present

MIT Lincoln Laboratory

Technical Staff, Applied Space Systems Group

Oct. 2022 – July 2024

Lexington, MA

- Performed modeling campaign of ionosphere on parallel computing grid, developing software pipeline to compare different models, and assimilating various sources of data. Developed rigorous and relevant metrics to evaluate model performance against data.
- Determined sources of astrometric errors and anomalies in TESS asteroid search analysis.
- Evaluated instrument proposals for NASA/NOAA GOES/GeoXO weather satellite program to ensure that they met government requirements. Modeled electro-optical system performance.

University of Washington

Graduate Research Assistant, Muon g-2 experiment, Advisor: David Hertzog

Jul. 2016 – Dec. 2022

Seattle, WA

- Led University of Washington effort for precession frequency analysis, key component of 2023 publication. Produced full set of systematic uncertainty estimates related to fitting procedure, and determined sources of unexpected behavior in data.
- Implemented improvement to algorithm to correctly reconstruct positron time and energy, resulting in a 4x decrease in mischaracterized events (pileup).
- Co-managed operation of 24 primary physics detectors (calorimeters) for experiment. Included on-site and on-call debugging and repairs, detector calibration, and startup and shutdown tasks.
- Took special systematics data and performed analysis to measure the lost muon systematic error, which was included as a correction to the published Run-1 result.
- Characterized, installed, and analyzed data from T0 detector to measure beam profiles and beam timing, essential for establishing the “start time” for each muon measurement period.
- Designed, built, and installed new laser monitoring detectors to ensure timing and energy stability of physics data. Mentored two undergraduate students in projects to analyze the data from this detector.

- Managed data quality monitor, website to provide real-time feedback on the data as it was collected. Implemented new pages and metrics to improve shifter experience.

University of Washington

Graduate Teaching Assistant

Sep. 2016 – Jun. 2017

Seattle, WA

- Led tutorial and lab sections for introductory physics courses.
- Provided tutoring to students in study center setting.

University of Washington

Graduate Research Assistant, ATLAS experiment at CERN, Advisor: Anna Goussion

Jun. 2016 – Sep. 2016

Geneva, Switzerland

- Worked on muon veto for muons faking taus.

University of Chicago

Undergraduate Research Assistant, ATLAS experiment at CERN, Advisor: Young-Kee Kim

Jan. 2014 – Jun. 2016

Chicago, IL

- Performed analysis of Monte Carlo simulation in C++ to study Higgs-bb channel

AWARDS

Rising Stars in Experimental Particle Physics

2021

- Selected as a Rising Star in Experimental Particle Physics by University of Chicago. Provided with honorarium for talk.

Miller Scholarship

2020

- Monetary prize awarded to an outstanding physics graduate student at the University of Washington, awarded after general exam.

Universities Research Association Visiting Scholars Program

2019

- Provided funding for travel to Fermilab for graduate student research.

First Year Excellence in Teaching

2018

Baumgartner Fellowship

2016

- Awarded to entering graduate students who show exceptional promise, providing a research fellowship for one quarter.

University Scholarship

2012-2016

- Merit scholarship awarded by the University of Chicago.

SELECTED TALKS

- “Status of the Muon g-2 experiment: Run-1 and beyond,” invited talk, Lake Louise Winter Institute (2022).
- “Precession frequency analysis in the Muon g-2 experiment: Run-1 and beyond”, invited talk, MIT Laboratory for Nuclear Science Special Seminar (2022).
- “Clustering improvements for the Run-2 and Run-3 precession frequency analysis in the Muon g-2 experiment,” contributed talk, American Physical Society April Meeting (2022).
- “Muon g-2 experiment: first results and future prospects,” invited talk, Rising Stars in Experimental Particle Physics Symposium, University of Chicago (2021).
- “A data-driven approach to the lost muon systematic uncertainty in the Muon g-2 experiment at Fermilab,” contributed talk, American Physical Society April Meeting (2020).

SELECTED PUBLICATIONS

- D. P. Aguillard et al. (Muon g-2 Collaboration), “Detailed report on the measurement of the positive muon anomalous magnetic moment to 0.2 ppm,” *Phys. Rev. D* **110**, 032009 (2024), doi: <https://doi.org/10.1103/PhysRevD.110.032009>.

- D. P. Aguillard et al. (Muon g-2 Collaboration), “Measurement of the Positive Muon Anomalous Magnetic Moment to 0.2 ppm,” *Phys. Rev. Lett.* **131**, 161802 (2023), doi:10.1103/PhysRevLett.131.161802.
- T. Albahri et al. (Muon g-2 Collaboration), “Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab,” *Phys. Rev. Accel. Beams* **24**, 044002 (2021), doi:10.1103/physrevaccelbeams.24.044002.
- B. Abi et al. (Muon g-2 Collaboration), “Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm,” *Phys. Rev. Lett.* **126**, 141801 (2021), doi:10.1103/PhysRevLett.126.141801.
- K.S. Khaw, M. Bartolini, H. Binney, et al. (Muon g-2 Collaboration), “Performance of the Muon g-2 calorimeter and readout systems measured with test beam data,” *Nucl. Instrum. Methods Phys. Res., Sect. A* **945**, 162558 (2019), doi:10.1016/j.nima.2019.162558.

INTERNAL NOTES

- E989 Note 288: Energy-dependent time resolution for reconstruction (2022).
- E989 Note 248: Measurement of the phase-momentum correlation lost muon systematic error (2020).

OUTREACH

- Interviewed for New York Times article “Physicists move one step closer to a theoretical showdown” (2023).
- “What a big magnet can tell us about particle physics,” outreach talk to Girls Rock Science and Math (3rd-5th graders) (2021).
- Interviewed for Science Magazine article “Particle physics mystery deepens, as physicists confirm that the muon is more magnetic than predicted” (2021).
- Invited panelist for “How to succeed in graduate school” panel at Conference for Undergraduate Women in Physics, University of Washington (2019).
- Invited panelist for “Diversity and inclusion” panel at New Perspectives Conference, Fermilab (2018).